

Lecture 2: Computer System, Basic Components

Learning Objectives

By the end of this lecture, students should be able to:

- Identify the major components of a computer system.
 - Explain the role of the CPU.
 - Understand the ALU and Control Unit.
 - Explain the purpose of registers.
 - Understand primary and secondary memory.
 - Explain input and output devices.
 - Understand the concept of system buses.
 - Describe how these components work together when executing a program.
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1. Introduction

Teaching Script

Hello everyone, and welcome back to our **Computer Organization and Assembly Language** course.

In our previous lecture, we discussed what Computer Organization and Computer Architecture are, and we also introduced Assembly Language and Machine Language.

Today, we are going to take a closer look at the basic components that make up a computer system.

Whenever we use a computer, we usually see things like the keyboard, mouse, monitor, and CPU.

But internally, a computer contains many different components, and each component has a specific job.

So today's question is:

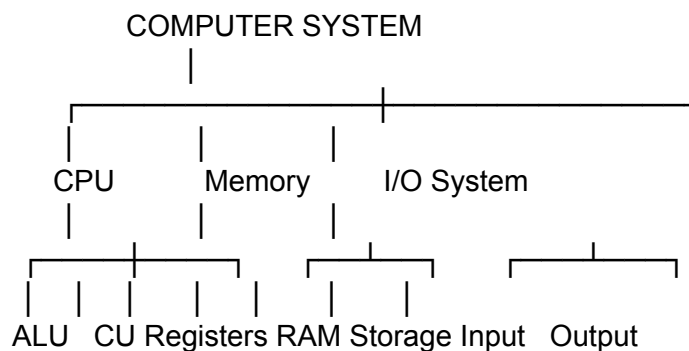
What are the major components of a computer system, and how do they work together?

By the end of this lecture, you should have a clear picture of how data moves through a computer and how the CPU, memory, input/output devices, and buses communicate with each other.

So let's begin.

2. Basic Components of a Computer

A computer system can be divided into several major components:



The major components are:

1. **CPU**
2. **Memory**
3. **Input devices**
4. **Output devices**
5. **Storage**
6. **System buses**

Let's understand each one.

3. Central Processing Unit — CPU

The **CPU**, or **Central Processing Unit**, is often called the **brain of the computer**.

Its primary job is to:

- Fetch instructions
- Decode instructions
- Execute instructions
- Process data
- Control operations

In simple words:

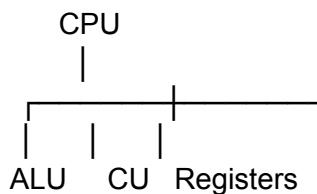
The CPU executes the instructions of a program.

For example, suppose our program says:

5 + 10

The CPU needs to process the instructions required to perform this operation.

The CPU contains several important components.



The three important components we'll focus on are:

- **ALU**
- **Control Unit**
- **Registers**

4. Arithmetic Logic Unit — ALU

The **ALU**, or **Arithmetic Logic Unit**, performs arithmetic and logical operations.

Arithmetic operations

For example:

Addition

Subtraction

Multiplication

Division

Logical operations

For example:

AND
OR
NOT
XOR

It can also perform comparisons such as:

$5 > 3$
 $10 == 10$
 $7 < 2$

Example

Suppose the CPU needs to calculate:

$10 + 20$

The ALU performs the arithmetic operation.

$10 + 20$
↓
ALU
↓
30



Teaching Explanation

Think of the ALU as the calculator inside the CPU.

Whenever the processor needs to perform an arithmetic or logical operation, the ALU is responsible for performing that operation.

However, the ALU doesn't decide by itself what operation it should perform.

Something needs to tell the ALU what to do.

That is where the Control Unit comes in.

5. Control Unit — CU

The **Control Unit** controls and coordinates the activities of the CPU.

It tells different components:

- What operation to perform
- When to perform it
- Where data should come from
- Where results should go

Think of the Control Unit as the **manager or coordinator** of the CPU.

For example:

Instruction:
ADD 5 + 10

↓

Control Unit

↓

"ALU, perform addition"

↓

ALU

↓

Result

The Control Unit doesn't normally perform the arithmetic itself.

Instead, it generates the necessary control signals that coordinate the operation.

6. Registers

Registers are **small, extremely fast storage locations inside the CPU**.

They temporarily hold:

- Data
- Instructions
- Addresses
- Intermediate results

For example, imagine:

Register A = 10

Register B = 20

The ALU can use these values:

10 + 20

↓

ALU

↓

30

The result may then be stored in another register.

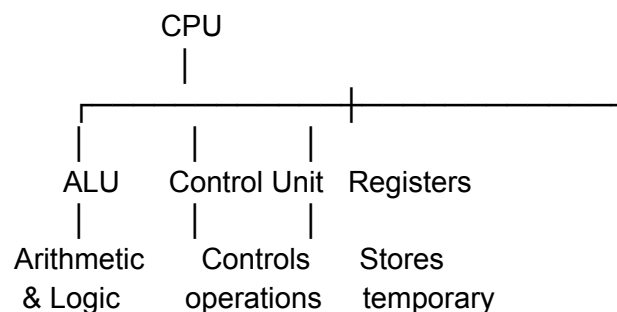
Why do we need registers?

Because the CPU needs extremely fast access to the data it is currently processing.

Registers are much smaller than RAM, but they are extremely fast.

7. CPU Summary

At this point, students should understand:



Operations data

Easy way to remember

ALU → **Calculates**

Control Unit → **Controls**

Registers → **Temporarily Store**

8. Memory

Now move outside the CPU.

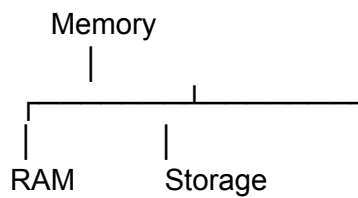
The second major component is **memory**.

Memory stores:

- Programs
- Instructions
- Data
- Intermediate results

When you open an application, the operating system loads the required program and data into memory so the CPU can access them.

A simplified view is:



We need to distinguish between these two.

9. RAM — Random Access Memory

RAM is the main memory used by a computer while programs are running.

For example, when you open:

- Chrome
- VS Code
- Microsoft Word
- A game

the required program data is loaded into RAM.

Important characteristics

RAM is:

- Fast
- Read/write
- Volatile

What does volatile mean?

It means that RAM loses its stored data when power is removed.

For example:

Computer ON

↓

RAM contains data

Computer OFF

↓

RAM loses its contents

10. Secondary Storage

Secondary storage is used for **long-term data storage**.

Examples include:

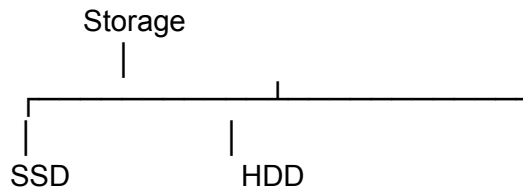
- SSD
- HDD
- USB drives

- Memory cards

Unlike RAM, storage is generally **non-volatile**.

That means data remains stored even after the computer is turned off.

For example:



11. RAM vs Storage

This is an important exam question.

RAM	Storage
Temporary working memory	Long-term storage
Volatile	Non-volatile
Very fast	Generally slower than RAM
Used by running programs	Stores files and programs
Usually smaller	Usually larger

Example

Suppose you install a game.

The game might occupy:

100 GB

on your SSD.

When you launch it, some of the game's required data is loaded into RAM.

So:

SSD



Game stored permanently

RAM



Game data currently needed

CPU



Executes instructions

12. Input Devices

Input devices allow us to **send data or commands to the computer**.

Examples:

- Keyboard
- Mouse
- Microphone
- Scanner
- Camera
- Touchscreen

For example:

When you type:

Hello

using a keyboard, the keyboard sends input to the computer.

Conceptually:

Keyboard



Computer



13. Output Devices

Output devices present processed information to the user.

Examples:

- Monitor
- Printer
- Speakers
- Projector

For example:

Keyboard
↓
Computer
↓
CPU processes data
↓
Monitor
↓
User sees result

14. Input → Processing → Output

This is one of the simplest ways to understand how a computer works.

INPUT
↓
PROCESSING
↓
OUTPUT

For example, imagine a calculator.

You enter:

10 + 20

Step 1 — Input

You enter the numbers using the keyboard.

Step 2 — Processing

The CPU processes the operation.

Step 3 — Output

The result:

30

is displayed on the screen.

15. System Buses

Now introduce an important COAL concept.

Different computer components need to communicate with one another.

For example:

CPU ↔ Memory

CPU ↔ I/O

Memory ↔ I/O

The communication takes place through **buses**.

A bus is essentially a communication pathway used to transfer information between components.

The three major system buses are:

1. **Data Bus**

2. **Address Bus**
 3. **Control Bus**
-

16. Data Bus

The **Data Bus** carries actual data.

For example:

CPU ←—— Data ——→ Memory

If the CPU wants to read a value from memory, the data is transferred through the data bus.

Think:

Data Bus = What data is being transferred?

17. Address Bus

The **Address Bus** carries the address of the memory location that the CPU wants to access.

For example:

CPU
|
| Address = 1000
↓
Memory Location 1000

Think:

Address Bus = Where is the data?

18. Control Bus

The **Control Bus** carries control signals.

These signals tell components what operation should happen.

For example:

READ
WRITE
INTERRUPT
CLOCK

Think:

Control Bus = What should happen?

19. Three Buses — Easy Trick

Give your students this simple formula:

Data Bus

WHAT?

Address Bus

WHERE?

Control Bus

HOW/WHEN?

For example:

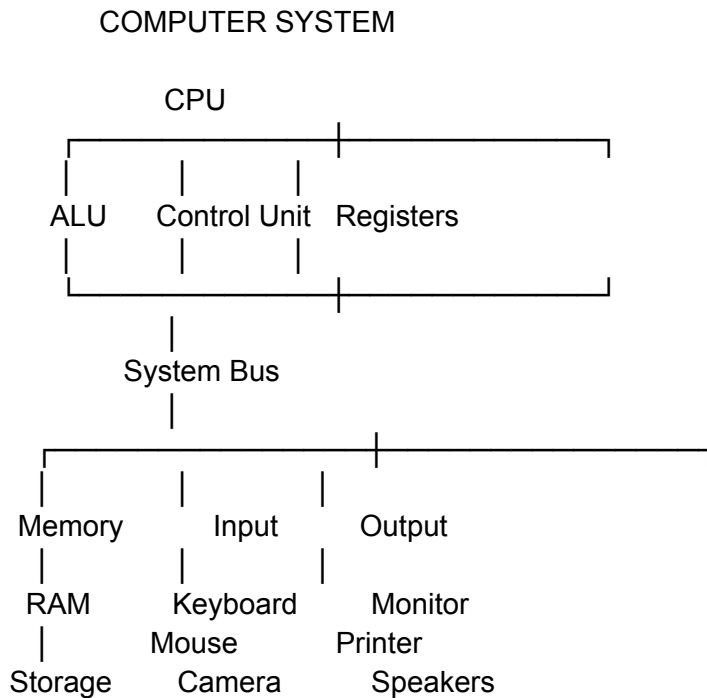
Data Bus → 50
Address Bus → 1000
Control Bus → READ

Meaning:

"Read the data from memory address 1000."

20. Complete Computer System

Now combine everything you've taught.



Explain that this is a simplified conceptual model. Modern computer systems are considerably more complex.

21. Real-Life Example — Opening a Program

Now give students a complete example.

Suppose you click **Google Chrome**.

Step 1 — Input

You click the Chrome icon using your mouse.

Mouse → Computer

Step 2 — Operating System

The operating system receives the request and loads the necessary program data.

Step 3 — Storage

Chrome's executable files are stored on your SSD.

SSD → RAM

Step 4 — Memory

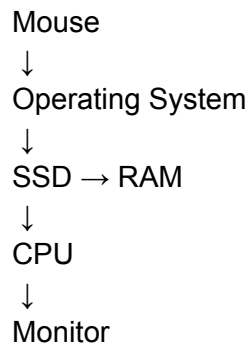
The required program data is loaded into RAM.

Step 5 — CPU

The CPU fetches and executes instructions.

Step 6 — Output

The result appears on your monitor.



This is a simplified view, but it gives students the correct overall idea.

22. Example — Adding Two Numbers

Let's connect this to assembly language.

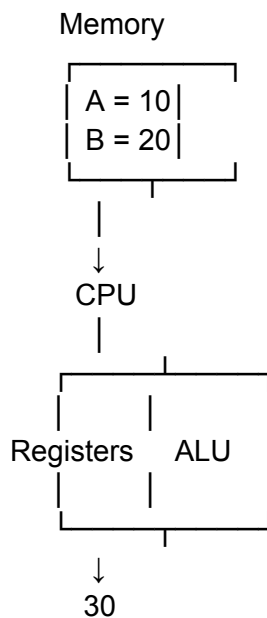
Suppose we want:

$A = 10$

$B = 20$

$C = A + B$

Conceptually:



The actual assembly instructions will depend on the architecture.

For MIPS, which we will study later, the idea might involve loading values into registers and then using an **add** instruction.

We will not worry about the exact MIPS code yet.

That will come later.

Lecture Summary

 Closing Script

So, let's quickly summarize what we learned in today's lecture.

A computer system consists of several major components that work together.

The **CPU** is responsible for executing instructions.

Inside the CPU, the **ALU** performs arithmetic and logical operations, the **Control Unit** coordinates operations, and **Registers** provide extremely fast temporary storage.

RAM stores programs and data that are currently being used, while secondary storage such as SSDs and HDDs provides long-term storage.

Input devices allow us to provide data to the computer, while **output devices** present the results to us.

Finally, we discussed **system buses**.

The **Data Bus** carries data.

The **Address Bus** carries addresses.

And the **Control Bus** carries control signals.

In the next lecture, we're going to build on these concepts and discuss **Computer Architecture vs Computer Organization** in much greater detail.

We'll understand exactly what the programmer can see, what is hidden inside the hardware, and why two computers can have the same architecture but completely different internal organizations.

Thank you for watching, and I'll see you in the next lecture.
